CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT) SEM - II University Examination, 2010-11

MA - 102 Engineering Mathematics-II

Date: 09.05.11, Monday Time: 1.30 p. m to 4.30 p. m. Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

(4) Rough work is done in the last page of main answer book, don't write anything on the question paper.

SECTION-I

Q-1 (a) Solve by the method of variation of parameters : $\frac{d^2y}{dx^2}+4y=\sec 2x$. [05]

(b) Solve : $(x^2+y^2-a^2)x\,dx+(x^2-y^2-b^2)y\,dy=0$. [03]

(c) Solve : $p+q=\sin x+\sin y$. [03]

Q-2 (a) Solve : $\frac{dy}{dx}+\frac{4x}{x^2+1}y=\frac{1}{(x^2+1)^3}$. [04]

(b) Solve: (i) $8y''-2y'-y=0$ (ii) $(D^2+2kD+k^2)y=0$. [04]

(c) Solve the initial-value problem: $y''+y'=2+2x+x^2$, $y(0)=8, y'(0)=-1$. [04]

OR

Q-2 (a) Solve: $\frac{dz}{dx}+\frac{z}{x}\log z=\frac{z}{x^2}(\log z)^2$. [04]

(b) Solve: $y'''+3y''+3y'+y=8e^x+x+3$. [04]

(c) Solve the initial-value problem: $4x^2y''+24xy'+25y=0$, $y(1)=2, y'(1)=-6$. [04]

Q-3 (a) Solve $(mz-ny)\frac{\partial z}{\partial x}+(nx-lz)\frac{\partial z}{\partial y}=ly-mx$. [04]

(b) Solve $z^2\left(p^2+q^2+1\right)=1$. [04]

(c) Solve (i) $p+q+pq=0$ (ii) $p-x^2=q+y^2$. [04]

OR

Q-3 (a) Solve $(y-z)p+z(x-y)q=z-x$. [04]

(b) Solve $y^2p-xyq=x(z-2y)$. [04]

(c) Solve (i) $z-px-qy=2\sqrt{pq}$ (ii) $z=p^2+q^2$. [04]

SECTION-II

Q-4 (a) Prove $J_{1/2}(x)=\sqrt{\frac{2}{\pi x}}\sin x.$ **[03]**

(b) Change the order of integration in the integral $\int_0^\infty\int_0^\infty \frac{e^{-y}}{y}\,dy\,dx$ and evaluate it. **[03]**

(c) Find all the Eigenvalues and Eigenvectors of the matrix $A=\begin{bmatrix}2 & 2 & 1\\ 1 & 3 & 1\\ 1 & 2 & 2\end{bmatrix}$. **[05]**

Q-5 (a) Find the volume common to the cylinder $x^2+y^2=a^2$ and $x^2+z^2=a^2$. **[04]**

(b) If $A=\begin{bmatrix}-1 & 2+i & 5-3i\\ 2-i & 7 & 5i\\ 5+3i & -5i & 2\end{bmatrix}$, Show that A is a Hermitian matrix and iA is a skew Hermitian matrix. **[04]**

(c) Evalate $\iint r^3\,dr\,d\theta$ over the area included between the circles $r=2\sin\theta$ and $r=4\sin\theta$. **[04]**

OR

Q-5 (a) Evaluate $\iint_R dA$; R is the region bounded by $x=0, x=1, y=x^2, y=2-x$. **[03]**

(b) Find the characteristic equation of the matrix $A=\begin{bmatrix}2 & 1 & 1\\ 0 & 1 & 0\\ 1 & 1 & 2\end{bmatrix}$ Verify the Cayley – Hamilton theorem and hence evaluate the matrix equation $A^8-5A^7+7A^6-3A^5+A^4-5A^3+8A^2-2A+I$. **[05]**

(c) By changing into polar co-ordinates, evaluate the integral

$$\int_0^{2a}\int_0^{\sqrt{2ax-x^2}}(x^2+y^2)\,dx\,dy.$$ **[04]**

Q-6 (a) Express $f(x)=4x^3-3x^2+2x+1$ in terms of Legendre polynomials. **[04]**

(b) Adopting standard notations, Prove that $\frac{d}{dx}\left[x^n J_n(x)\right]=x^n J_{n-1}(x)$. **[04]**

(c) Evaluate $\int_0^1 x^m(\log\frac{1}{x})^n\,dx.$ **[04]**

OR

Q-6 (a) Prove $erf_c(x)+erf_c(-x)=2.$ **[03]**

(b) Evaluate (i) $\int_0^n x^n(n-x)^p\,dx$ (ii) $\int_0^\infty \frac{x^c}{c^x}\,dx,(c>1).$ **[05]**

(c) Prove (i) $P_n(1)=1$ (ii) $P_n(-x)=(-1)^n P_n(x)$. **[04]**